

Enhancing Mining Operations with Advanced Video Analytics



Situation

One of the world's leading mining companies in North America, wanted have strategies that can be implemented to minimize the operations disruptions at the dump pocket jams (DPJ) caused by wet muck and mud rush events in ore feeders, utilizing advanced video analytics, engineered mud gates and traditional sensors in real-time.



Action

Project is a multi-pronged approach to reduce the significant unplanned downtime attributed to wet muck and mud rush events. The project leverages uses high definition cameras with GPU's and analytical models are deployed on the edge devices. and traditional sensors to determine the trending and instantaneous risk state from a smart camera located at the exit of an ore feeder. The project will notify the operator through a dashboard of the telemetry device alert and interface with the control network to reduce the feeder speed or engage the feeder interlock if needed.



Impact

- Advanced Video Analytics, in conjunction with Deep Learning, have been shown to be extremely strong indicators of DPJs by capturing 95% of potential jams at an average of 8 mins before operator action
- Provides a flexible, low-cost solution with better sensitivity. Is easy to configure and much more accurate. Edge computing provides data processing and solution deployment on site, or at the equipment itself, enabling rapid response, reliability, and scalability

